

Topic: Memory Segmentation in 8086

Q. Explain the concept of Memory Segmentation in the 8086 microprocessor. How is a 20-bit physical address calculated using a 16-bit segment register and a 16-bit offset?

> **Definition to Remember**

Memory Segmentation in 8086

The CPU manages these memory blocks using four specialized 16-bit Segment Registers:

> **Me**
intern
genera

1. **C**

2. Logical vs Physical Address

- **Logical Address:** Format used by programmers `(Segment : Offset)`. e.g., `1000H:0002H`.

- **P**

3. Calculating the 20-bit Physical Address

The Bus Interface Unit (BIU) performs this calculation:

Form

Calculation Example:

- **CS**

1. **Shift Segment by 10H:** `2000H 10H` = `20000H` (20-bit Base)

2. **Ac**

4. Advantages of Segmentation

- **Code Relocation:** Programs can be moved anywhere in memory by changing only the Segment Register; internal offset addresses remain unchanged.

- **M**
data o

> **Must-Write Points (for 10 marks)**

> 1. 80

> **Quick Recall**

> `Seg
Offset